

PurpleAir  
live sensors

Open-Meteo  
weather

NOAA HMS  
smoke

CAMS  
AOD, PM2.5

EJScreen,  
census tracts

30-feature pipeline  
(BallTree neighbor PM2.5 at 25/50/100 km)

**RF + LightGBM + CatBoost simplex ensemble**  
**(LOSO  $R^2 = 0.713$ , 310 sensors)**

live tract map  
6,896 TX census tracts

**exact SHAP blend**  
**7 concept groups (this paper)**